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RESEARCH ARTICLE

Chest X-Ray Report Generation Using Abnormality Guided Vision Language Model

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ABSTRACT Automated radiology report generation is essential for assisting radiologists in managing the growing volume of chest radiographs. Unlike conventional image captioning models, radiology report generation must prioritize abnormality-specific feature extraction over generic image features. However, existing models lack an inherent mechanism to extract clinically significant abnormality features directly from radiographs without relying on external knowledge bases or auxiliary models. Additionally, most existing approaches use a single vision encoder, limiting their ability to capture complementary visual cues essential for accurate abnormality identification. To address these limitations, we introduce META-CXR (Multimodal Expert Tokens-based VLM for Abnormality-Guided Chest X-ray Reporting), a vision-language model (VLM) designed to enhance both abnormality classification and radiology report generation. META-CXR employs a multi-encoder visual backbone that combines CNNs, ViTs, and Swin Transformers to extract diverse and complementary visual features. These multi-encoder features serve as a shared representation and are utilized in two key components of the architecture: the META-Former module, a modified Q-Former designed to fuse heterogeneous encoder outputs into a unified token space for effective report generation, and the Multi-Head Cross Attention Classification (MHCAC) module, which performs multi-class, multi-label abnormality classification, including an explicit "uncertain" category. The identified abnormalities are then integrated into the language modeling process as soft prompts to ensure clinically coherent and diagnosis-aware report generation using a large language model (LLM). META-CXR achieves state-of-the-art performance, demonstrating an F1-score of 0.699 for cross-domain multi-class, multi-label abnormality classification on the CheXpert dataset. Additionally, it achieves strong natural language generation results, with a BERTScore of 0.426 and a METEOR score of 0.173 on the MIMIC-CXR test set. Ablation studies validate the contributions of the multi-encoder backbone, hierarchical classification via MHCAC, the META-Former fusion strategy, and classification-aware report generation. Furthermore, attention map visualizations improve interpretability by highlighting clinically relevant image regions, offering radiologists transparent insights into model decisions. By addressing key challenges in abnormality-specific feature extraction, uncertainty-aware classification, and diagnosis-driven reporting, META-CXR establishes a new benchmark for vision-language models in radiology AI.

INDEX TERMS Artificial intelligence, abnormality classification, chest X-ray reporting, computer vision, deep learning, multi-modal, vision-language model.

I. INTRODUCTION

Automated radiology report generation is increasingly critical in modern healthcare, particularly for chest radiographs. The growing demand for timely and accurate interpretation of chest X-rays, especially in high-throughput or

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resource-constrained settings, has placed significant strain on radiology departments. This makes intelligent, scalable solutions essential not only for improving clinical efficiency but also for ensuring consistent diagnostic quality across diverse patient populations. Unlike conventional image captioning models [1], [2], [3], which aim to generate descriptive captions by analyzing the entire image, radiology report generation presents a unique challenge where models must prioritize abnormality-specific feature extraction over generic image features to produce clinically relevant and comprehensive reports [4], [5]. Given that radiology reports are fundamentally derived from abnormality findings, effective report generation necessitates mechanisms capable of accurately identifying and describing pathological features within medical images.

Most recent approaches attempt to address this challenge by integrating medical knowledge through pre-built knowledge bases [4], [6], [7]. While knowledge augmentation can enhance model performance, maintaining and updating knowledge bases is often tedious and labor-intensive, which limits their scalability in real-world clinical workflows [8]. Moreover, many existing abnormality classification models rely solely on global visual feature extractors followed by conventional classification heads, lacking structured mechanisms to extract and prioritize multi-class, multi-label abnormality-specific features. This can lead to diagnostic inconsistencies, overconfidence in uncertain cases, and poor alignment between predicted findings and generated reports. Additionally, ensemble models combine the strengths of multiple deep learning architectures to improve accuracy, robustness, and reliability by leveraging diverse features and reducing the risk of incorrect predictions [9].

Recent advancements in large language models (LLMs) have demonstrated exceptional capabilities in contextual understanding, facilitating their integration with vision models via a range of multimodal fusion strategies. [10], [11]. As a result, vision-language models (VLMs) have been widely adopted in medical applications, such as radiology report generation and visual question answering [12], [13], [14], [15], [16], [17]. However, most existing medical VLMs rely on a single vision encoder, typically a Convolutional Neural Network (CNN) [18] or Vision Transformers (ViTs) [19], [20] to capture visual features, which restricts their ability to extract a rich and complementary range of visual features necessary for accurate diagnosis. Although studies like BRAVE [21] have explored combining multiple ViT-based encoders to enhance representational capacity, the integration of fundamentally different encoder families (CNNs, ViTs, Swin [22]) for abnormality classification and report generation remains underexplored.

To this end, we identify two key challenges in current medical VLMs:

- The absence of an in-built mechanism to extract abnormality-specific visual features for multi-class multi-label abnormality classification, including uncertain class.
- Reliance on a single vision encoder, which restricts the capacity to capture complementary visual features—an issue that can be mitigated by incorporating multiple vision encoders.

To address these challenges, we propose META-CXR (Multimodal Expert Tokens-based VLM for Abnormality-Guided Chest X-ray Reporting), a unified medical VLM designed to enhance both abnormality classification and radiology report generation through an in-built abnormality extraction mechanism. META-CXR employs a multi-encoder visual backbone that combines CNNs, ViTs, and Swin Transformers to leverage complementary inductive biases across architectural families. This enables the model to simultaneously capture fine-grained details(via CNN), global contextual cues(via ViT), and hierarchical spatial structures(via Swin) from radiographs. Building upon this rich visual representation, the proposed Multi-Head Cross-Attention Classification (MHCAC) module enables precise detection and classification of 14 distinct abnormalities through a hierarchical feature extraction mechanism. Additionally, it introduces an explicit "uncertain" category, addressing a critical yet underexplored challenge in medical AI—recognizing ambiguous cases where confident classification is not possible. This uncertainty-aware classification reduces overconfident misdiagnoses, making the model better aligned with real-world radiology workflows.

Furthermore, the rich multi-encoder representation is passed to a modified Q-Former [10], referred to as META-Former, which is specifically adapted to handle and align heterogeneous features from multiple vision encoders. META-Former fuses these representations into a unified learnable token space that is semantically structured and optimized for the downstream report generation task.

To ensure clinically relevant report generation, META-CXR directly integrates classification results from the MHCAC module into the report generation pipeline. The detected abnormalities are seamlessly incorporated into the report generation process via a soft prompt, along with the learned tokens from META-Former, which are then fed into a large language model (LLM). This classification-aware report generation ensures that the generated reports explicitly reflect the detected abnormalities, enhancing clinical coherence and diagnostic accuracy.

With the growth of techniques that interface with the medical domain, explainability and interpretability in medical image analysis are crucial for ensuring transparency, building trust among clinicians and patients, enabling effective clinical decision-making, and complying with regulatory requirements by allowing human experts to understand, validate, and act confidently on AI-driven diagnostic results [23], [24], [25], [26]. Therefore, beyond the robust classification and report generation capabilities, META-CXR prioritizes explainability through attention map visualizations, which highlight critical image regions that contribute to abnormality identification and report generation. This enhances clinical interpretability, allowing

radiologists to validate AI-generated insights with greater confidence.

META-CXR is extensively evaluated on the MIMIC-CXR [27], IU X-Ray [28], and CheXpert [29] test sets, achieving state-of-the-art performance in both multi-class multi-label classification and radiology report generation. Ablation studies validate the impact of hierarchical feature extraction and the multi-encoder setup, confirming their role in enhancing both classification accuracy and report coherence.

The key contributions of this paper are as follows:

- A multi-encoder visual backbone (CNN, ViT, Swin) to capture diverse and complementary visual features, along with META-Former—a dedicated fusion module designed to align and integrate these heterogeneous representations into a unified token space.
- Proposal of the MHCAC module, which leverages learnable expert tokens to refine abnormality-specific features, integrating uncertainty-aware classification to improve diagnostic reliability.
- Seamless integration of classification results into report generation, ensuring clinically coherent and diagnostically relevant reports through classification-aware report generation.
- Comprehensive evaluation on the MIMIC-CXR, IU X-Ray, and CheXpert datasets demonstrates state-of-the-art performance in both abnormality classification and report generation, further validated through ablation studies and expert assessments.

The paper is structured as follows: Section II reviews the background and related work on multi-modal learning, vision-language models, and radiology report generation. Section III details the proposed methodology, including the META-CXR architecture, the MHCAC module, and training procedures. Section IV presents the experimental results and ablation studies, while Section V discusses the key findings and expert assessment. Finally, Section VI concludes the paper by summarizing the contributions, limitations, and potential directions for future research.

II. BACKGROUND

A. RELATED STUDIES

1) VISION-LANGUAGE MODELS (VLM) IN MEDICINE

Recently, VLMs have demonstrated great potential in tasks that require the integration of visual data and text data, specifically tasks like radiology report generation. Most of the medical VLMs often leverage pre-trained domain-specific vision and language models to fuse image features with linguistic representations, enabling more accurate report generation.

a: VISION ENCODERS IN VLMs

Vision encoders form the backbone of vision-language models (VLMs), directly influencing the quality of visual representations used in the generation phase. Convolutional Neural Networks (CNNs), such as ResNet [30], are widely adopted in medical imaging for their strong inductive biases toward local texture patterns and spatial hierarchies, enabling effective learning even with limited data. On the other hand, Vision Transformers (ViTs) offer a complementary perspective by capturing long-range dependencies and global contextual information through self-attention mechanisms [19]. Swin Transformers further enhance this by introducing hierarchical processing and local-windowed attention, allowing scalable modeling of both global and local features [22]. These architectural families bring distinct inductive biases: CNNs favor localized feature extraction, ViTs excel at capturing global semantics, and Swin Transformers balance both through hierarchical token aggregation.

Recent efforts like BRAVE [21] have explored the combination of different ViT variants to harness complementary visual cues, demonstrating improved visual representations in VLMs. However, many existing medical VLMs [17], [31], [32], [33] rely on a single vision encoder, thereby limiting the diversity and richness of visual representations. In contrast, META-CXR integrates three diverse encoder families—CNNs, ViTs, and Swin Transformers—to unify their complementary strengths. This fusion captures fundamentally different inductive biases across architectural paradigms, enabling the construction of a more comprehensive and semantically rich visual representation that better supports report generation in complex clinical scenarios.

b: MULTI-MODAL FUSION IN VLMs

Multi-modal fusion plays a pivotal role in vision-language models (VLMs), where the effective alignment of visual and textual modalities is essential for generating coherent and clinically meaningful radiology reports. Recent advances, such as BLIP-2 [10], employ multimodal transformers with cross-attention mechanisms to bridge the modality gap. In particular, the Q-Former architecture introduced in BLIP-2 facilitates efficient vision-language alignment, which has demonstrated improvements in report generation performance [16], [17]. Flamingo [11] further advanced this concept by interleaving visual inputs within frozen language model blocks to enable iterative multimodal reasoning; this has since been extended to the medical domain in Med-Flamingo [34] and Flamingo-CXR [35]. More recently, BRAVE [21] introduced the MEQ-Former module, which integrates representations from multiple vision encoders, offering an enriched visual context for language modeling. However, its fusion mechanism is restricted to ViT-based encoders. In contrast, META-CXR generalizes this idea by extending fusion across fundamentally different vision encoder families—CNNs, ViTs, and Swin Transformers—each contributing unique inductive biases. The resulting module, META-Former, enables a more expressive and semantically structured vision-language fusion strategy, tailored for the complexity of clinical tasks.

2) RADIOLOGY REPORT GENERATION (RRG)

Radiology report generation has emerged as a critical research area due to the increasing need for automated systems that can assist radiologists in handling large volumes of medical imaging data. Early models approached the task from a captioning perspective [1], [2], [3], [36], converting images to text. However, these methods often failed to capture the complexity and clinical relevance required in radiology reports. More recent advancements have introduced multi-modal models that aim to generate structured and contextually accurate reports by integrating both visual and textual information [4], [37].

a: MEDICAL KNOWLEDGE INTEGRATION IN RRG

Recent studies have increasingly focused on incorporating medical knowledge to enhance radiology report generation [4], [6], [7], [38], [39], [40], [41]. For instance, [4] utilized fine-grained image-text alignment enhanced with disease knowledge, leveraging knowledge graphs to model relationships between diseases, abnormalities, and anatomical terms, ensuring clinical and visual alignment in the reports. Other frameworks [6], [17], [42], [43] employ abnormality information derived from visual features as prompts to guide report generation. Additionally, [44] proposed detecting abnormal anatomical regions and generating descriptive sentences for each region. These approaches emphasize isolating critical regions and integrating structured medical knowledge, such as abnormality classification outputs and knowledge graphs, to improve the clinical relevance and quality of the generated reports.

b: STRUCTURED ABNORMALITY CLASSIFICATION FOR ENHANCED RRG

Motivated by the fact that medical knowledge can guide report generation, the accurate classification of abnormalities in radiographs has a significant impact on generating meaningful and relevant reports. Existing multi-label classification methods proposed in works like [6], [17], [42], [45] predominantly rely on unimodal visual feature extractors such as ResNet or ViTs, which lack structured mechanisms for fine-grain abnormality classification. Additionally, these methods do not model uncertainty, leading to overconfident predictions in ambiguous cases. META-CXR addresses these challenges by introducing a hierarchical feature extraction-based classification mechanism utilizing multiple vision encoders and explicitly incorporating an 'uncertain' category to enhance diagnostic reliability.

c: LARGE LANGUAGE MODELS (LLMS) FOR REPORT GENERATION

The emergence of large language models (LLMs), particularly those based on GPT-style architectures, has significantly advanced medical text generation. Despite their strong language modeling capabilities, general-purpose LLMs often underperform in domain-specific tasks like radiology report generation due to limited clinical knowledge and unfamiliarity with structured medical language.

To address these limitations, several domain-adapted vision-language models (VLMs) have been proposed. MedViLL [46] and X-REM [31] leverage BERT-based architectures for report decoding, while RepsNet [32] utilizes GPT-2 for autoregressive report generation. CXR-RePaiR-Gen [47] combines GPT-4 with retrieval-augmented generation (RAG) to reduce hallucinations. Other recent models such as LLaVA-Med [14], Med-Flamingo [34], Kargen [39], RaDialog [17], TRRG [43], and MAIRA-2 [33] build on LLaMa and Mistral-based language models, often incorporating clinical encoders like ClinicalBERT [48] or using auxiliary metadata (e.g., prior reports, indication sections) to enhance report accuracy. Among these, MAIRA-2 has explored grounded generation by linking textual findings to image regions using coordinate tokens or bounding boxes, aiming to improve interpretability and factual consistency.

Overall, fine-tuning LLMs on medical datasets and incorporating multimodal cues have consistently been shown to enhance the clinical fidelity of generated radiology reports.

Despite significant advancements, existing radiology report generation models still face critical limitations. These include a reliance on unimodal vision encoders, the absence of structured mechanisms for abnormality classification, and a lack of uncertainty modeling. To overcome these challenges, we introduce META-CXR, a medical vision-language model designed to enhance report generation. The following section provides a detailed overview of META-CXR's architecture and workflow.

III. METHODOLOGY

We propose a novel medical vision-language model (VLM), META-CXR, tailored specifically for chest X-ray report generation with an in-built abnormality classification module. As illustrated in Figure 1, META-CXR comprises four key components: (i) a set of frozen vision encoders, designed to extract complementary visual features from radiographs; (ii) the META-Former module, which fuses heterogeneous visual features and aligns them with language tokens; (iii) the Multi-Head Cross Attention Classification (MHCAC) module, which performs multi-class, multi-label abnormality detection using learnable expert tokens; and (iv) a frozen large language model (LLM) for radiology report generation.

A. DATA MATERIALS

1) MIMIC-CXR-JPG DATASET

For pre-training and experiments with META-CXR, we utilized the MIMIC-CXR-JPG dataset [49], [50], a large, publicly available collection of 377,110 chest X-ray images from 227,827 radiographic studies. As an improved version of the MIMIC-CXR dataset [27], MIMIC-CXR-JPG provides images in JPEG format, which simplifies storage and processing while facilitating medical computer vision

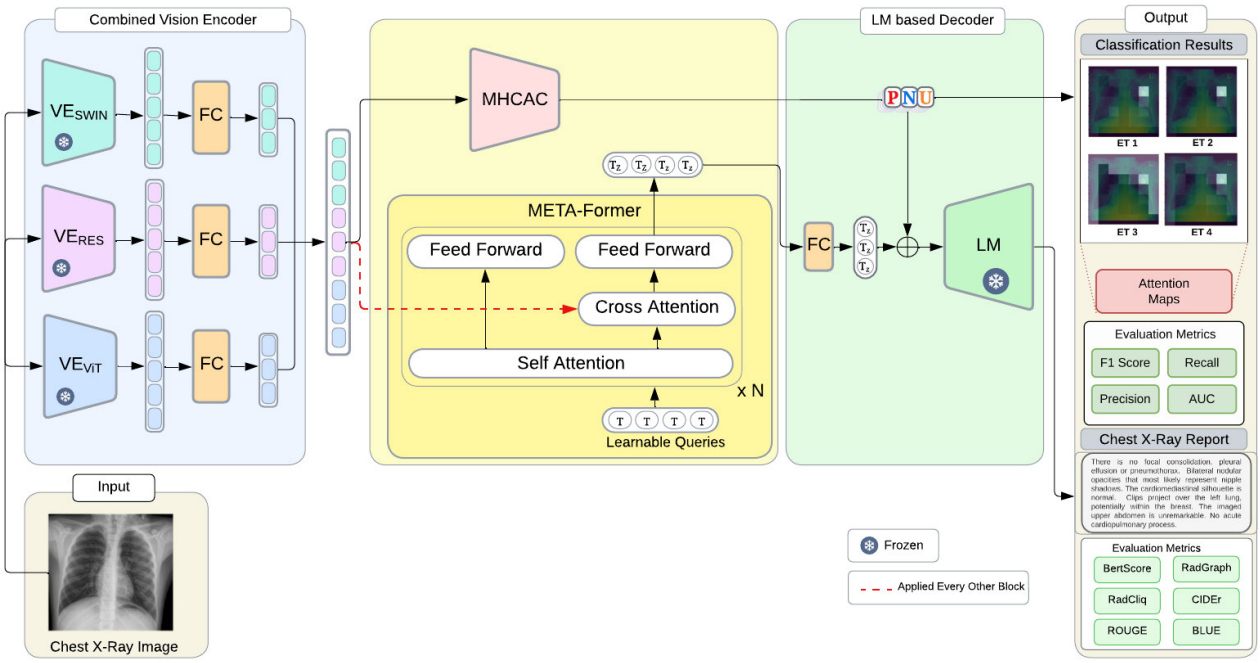


FIGURE 1. **META-CXR high level architecture.**

research. The dataset includes labels for 14 abnormality classes extracted from free-text radiology reports using natural language processing (NLP) tools like NegBio [51] and CheXpert labeler [29]. For our experiments, we used the CheXpert labeler, which applies a rule-based approach to classify mentions of medical terms into positive, negative, or uncertain labels. For our experiment, missing (NaN) values were treated as the negative class for consistency. The official dataset split is adapted, where it consists of 222758 training samples, 1808 validation samples, and 3269 testing samples. We focused exclusively on frontal views, as they provide standardized, diagnostically significant representations essential for disease classification and automated report generation.

The MIMIC-CXR-JPG dataset offers significant advantages over other radiology datasets. Its large size supports robust model training, while the use of standardized labels ensures consistency in analysis. Unlike DICOM formats, JPEG images simplify computational requirements without sacrificing usability. The dataset's rich metadata, such as view position and patient orientation, enables diverse research applications. Furthermore, its public availability and predefined data splits foster reproducibility and benchmarking across studies. Compared to other existing radiology datasets [52], MIMIC-CXR-JPG offers a more extensive collection, higher demographic diversity, and integration with electronic health records (EHR) through MIMIC-IV. These features establish MIMIC-CXR-JPG as a benchmark resource for advancing chest X-ray classification and related tasks [53].

TABLE 1. **Number of MIMIC-CXR-JPG images for each label (P: Positive, N: Negative, U: Uncertain).**

Class	Train			Test			Val		
	P	N	U	P	N	U	P	N	U
Atelectasis	48181	10759	1579	780	204	22	391	103	8
Cardiomegaly	46733	6242	16344	909	122	182	398	69	131
Consolidation	111287	4469	8343	227	108	99	88	52	57
Edema	28329	13725	26149	720	308	307	274	119	206
Enlarged Cardio	7462	9756	5442	168	216	69	64	106	52
Fracture	4680	586	951	96	5	3	18	3	6
Lung Lesion	6457	1163	845	120	18	13	70	10	7
Lung Opacity	53592	3961	3108	1099	79	56	411	29	18
No Finding	79148	0	0	583	0	0	632	0	0
Pleural Effusion	56536	6015	27718	1103	145	332	515	43	204
Pleural Other	2009	763	130	64	15	0	17	5	0
Pneumonia	17083	18926	24507	360	351	265	134	155	172
Pneumothorax	10988	1193	44292	101	20	761	83	10	379
Support Devices	71058	249	3628	1200	4	34	641	4	36

2) IU X-RAY DATASET

To assess the generalizability of META-CXR in diverse clinical settings, we additionally evaluate it on the Indiana University Chest X-ray Collection Dataset (IUXray) [28], [54]. It is the most widely used publicly accessible dataset in medical report generation tasks. It contains 3,955 fully de-identified radiology reports, each of which is associated with frontal and/or lateral chest X-ray images, and 7,470 chest X-ray images in total. Each report is comprised of several sections: Impression, Findings, Indication, etc. In this

work, we adopt the conventional data set partitioning as [37] for a fair comparison, with a train/test/val set of 7:1:2 of the entire dataset.

3) CheXpert DATASET

CheXpert [29], [55] is a large-scale dataset comprising 224,316 chest radiographs from 65,240 patients. It includes an automated labeler designed to detect the presence of 14 radiological observations while capturing the inherent uncertainties in radiograph interpretation. In our study, we leverage the CheXpert validation dataset to evaluate the cross-domain abnormality classification performance of the proposed MHCAC module, assessing its ability to generalize across diverse pathological conditions without explicit fine-tuning.

B. COMBINED VISUAL FEATURE EXTRACTOR

Visual feature extraction is performed using a set of frozen vision encoders. Motivated by the work of [21], we propose to use multiple vision encoders with different, yet potentially complementary, visual biases to create a more capable visual feature extractor. Since Transformers lack some of the inductive biases inherent to CNNs and therefore generalize poorly with insufficient and unbalanced data [19], we incorporate CNN, ViT, and Swin Transformer to enhance visual feature extraction. CNN-based encoders like ResNet are more adept at extracting local features [56], while ViTs are better suited to capture global features [18], [20]. As depicted in Figure 1, the patch-wise visual features from multiple encoders are then linearly projected to a shared embedding space and concatenated sequence-wise before being fed into META-Former and MHCAC.

C. META-FORMER: CROSS-MODAL FUSION ARCHITECTURE

META-Former, our multi-modal fusion mechanism, is a modified version of Q-Former [10], consisting of two transformer submodules: an image transformer and a text transformer. The inputs to META-Former are the concatenated visual features of the chest X-ray image from the combined visual encoder, corresponding radiology report tokens, and a set of learnable query tokens.

In META-Former, we combine Image-Text Contrastive (ITC) [10], Image-Grounded Text Generation (ITG) [10], and Image-Text Matching (ITM) [10] to achieve multimodal fusion for comprehensive chest X-ray report generation. ITC aligns image and text representations by maximizing mutual information, using a unimodal attention mask where queries and text tokens do not interact. This allows the model to learn global representations without direct cross-modal interaction. ITG uses a multimodal causal attention mask, enabling learnable queries to attend to each other and pass relevant visual information to text tokens, helping the model generate reports grounded in image data. ITM enables fine-grained alignment across modalities using a bidirectional self-attention mask,

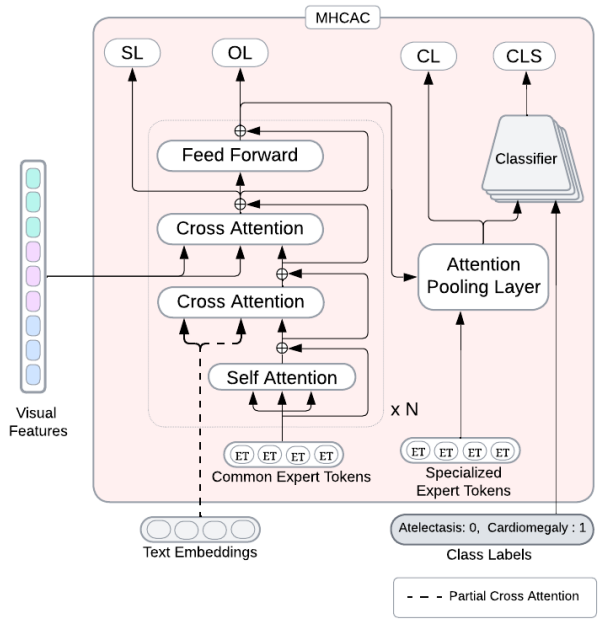


FIGURE 2. MHCAC module.

allowing queries and text tokens to attend to each other. This task ensures deep cross-modal interaction, enhancing the multimodal understanding of the image and report tokens through learnable query tokens. Overall, ITC, ITG, and ITM tasks optimize multi-modal alignment using distinct attention masks to enhance the integration of image and text features. The model consistently employs BLIP-2-inspired attention masking to enhance multimodal understanding and performance.

D. MHCAC: MULTI-HEAD CROSS-ATTENTION CLASSIFICATION MODULE

The MHCAC Module is designed to detect multiple abnormalities within a chest radiograph, each with three possible labels: positive, negative, and uncertain.

To extract abnormality-specific information, a set of common expert tokens and 14 specialized expert tokens are introduced, each corresponding to a specific abnormality. The common expert query tokens perform partial cross-attention with text embeddings associated with each image and cross-attention with concatenated image features to extract common features related to all abnormalities. These language-guided common visual features subsequently cross-attend with the 14 specialized expert tokens to pool abnormality-specific features through an attention-pooling layer. The pooled representations are then passed through independent classification heads, each of which outputs logits for three possible labels: positive, negative, and uncertain, which are converted into probabilities using the softmax activation function.

The MHCAC module is trained using Weighted Cross-Entropy Loss (CLS) as the primary objective function.

Additionally, Contrastive Loss (CL) [57] is introduced to enhance the separation between classes for each abnormality. Orthogonal Loss (OL) [58], [59] is applied to common expert tokens to ensure the extraction of distinct features, while Sparsity Loss (SL) [60] is introduced to encourage the attention weights to focus on the most relevant image patches. These combined loss functions collectively enable effective learning for the multi-class, multi-label classification task.

Here's how we can formalize the work done by the MHCAC module during pre-training stage.

1) INPUTS AND INITIALIZATIONS

The MHCAC module receives the following inputs:

- Image Embeddings: Feature representations extracted from three different vision encoders: CNN ($\mathbf{I}_1$), Vision Transformer (ViT) ($\mathbf{I}_2$), and Swin Transformer ($\mathbf{I}_3$).
- Text Embeddings: Linguistic representations ($\mathbf{T}$) derived from the corresponding medical reports.

The Common Expert Tokens ($\mathbf{E}_{\text{common}}$), which capture high-level abnormality features, and the Specialized Expert Tokens ($\mathbf{Q}$), which extract features specific to distinct abnormalities, are initialized randomly to allow adaptive learning during training.

2) EMBEDDING ALIGNMENT

The embedding alignment phase aligns embeddings from multiple vision encoders and text embeddings into a shared dimensional space of $d_c = 768$. Image embeddings from Vision Encoders 1, 2, and 3 are represented as $\mathbf{I}_1 \in \mathbb{R}^{B\times N_1\times d_1}$, $\mathbf{I}_2 \in \mathbb{R}^{B\times N_2\times d_2}$, and $\mathbf{I}_3 \in \mathbb{R}^{B\times N_3\times d_3}$, respectively, while text embeddings are given by $\mathbf{T} \in \mathbb{R}^{B\times N_t\times d_t}$. These embeddings are projected into a unified space and concatenated as given in (1):

$$\mathbf{I}_{\text{proj}} = \text{Concat}\left(\mathbf{I}_1\mathbf{W}_1, \mathbf{I}_2\mathbf{W}_2, \mathbf{I}_3\mathbf{W}_3\right), \quad \mathbf{T}_{\text{proj}} = \mathbf{T}\mathbf{W}_T, \tag{1}$$

where the projection matrices $\mathbf{W}_1, \mathbf{W}_2, \mathbf{W}_3 \in \mathbb{R}^{d_i\times d_c}$ and $\mathbf{W}_T \in \mathbb{R}^{d_t\times d_c}$ ensure dimensional consistency.

3) MHCA LAYERS: MULTI-HEAD CROSS-ATTENTION LAYERS

The *Multi-Head Cross-Attention (MHCA)* layers refine the common expert token representations, $\mathbf{E}_{\text{common}} \in \mathbb{R}^{B\times N_e\times d_c}$, by integrating information from both image patches and text embeddings. The refinement process is conducted through a sequence of attention mechanisms, ensuring expert tokens capture both generalized and abnormality-specific features.

a: SELF-ATTENTION AMONG EXPERT TOKENS

To encourage knowledge sharing among expert tokens, a self-attention mechanism is applied as given in (2):

$$\mathbf{E}_{\text{self}} = \text{LN}\Big(\mathbf{E}_{\text{common}} + \text{MHA}(\mathbf{E}_{\text{common}}, \mathbf{E}_{\text{common}}, \mathbf{E}_{\text{common}})\Big). \tag{2}$$

where:

$\text{MHA}(Q, K, V)$ denotes the multi-head attention operation, where Q (query), K (key), and V (value) are input tensors, and $\text{LN}(X)$ represents layer normalization, applied to stabilize learning.

b: EXPERT-TO-TEXT CROSS-ATTENTION WITH DROPOUT

To prevent excessive reliance on textual features, a dropout mask is applied to the text embeddings before cross-attention, as shown in (3). To extract abnormality-specific language features, cross-attention is performed between the common expert tokens and the dropout-regularized text embeddings, as given in (4).

$$\mathbf{T}_{\text{drop}} = \mathbf{T}_{\text{proj}} \odot \mathbf{M}, \quad \mathbf{M} \sim \text{Bernoulli}(1 - p_{\text{text}}), \tag{3}$$

$$\mathbf{E}_{\text{text}} = \text{LN}\Big(\mathbf{E}_{\text{self}} + \text{MHA}(\mathbf{E}_{\text{self}}, \mathbf{T}_{\text{drop}}, \mathbf{T}_{\text{drop}})\Big), \tag{4}$$

where:

$\mathbf{T}_{\text{drop}}$ represents the dropout-applied text embeddings before attention, $\mathbf{M}$ is a Bernoulli mask, where p_{text} is the dropout probability.

c: EXPERT-TO-IMAGE CROSS-ATTENTION

Then, cross-attention between the refined expert tokens and visual features is performed as given in (5)

$$\mathbf{E}' = \text{LN}\Big(\mathbf{E}_{\text{text}} + \text{MHA}(\mathbf{E}_{\text{text}}, \mathbf{I}_{\text{proj}}, \mathbf{I}_{\text{proj}})\Big). \tag{5}$$

In the META-CXR training pipeline, 6 MHCA layers are employed for feature extraction. Since textual information is unavailable during inference, text embeddings should serve only as a guiding mechanism during training to ensure that the learned representations retain the relevant knowledge extracted from the text. To achieve this, text guidance is primarily utilized in the initial MHCA layers, while the deeper MHCA layers progressively shift their focus toward image-derived features.

Attention Pooling: The *Attention Pooling* module extracts abnormality-specific representations from the refined common expert token representations using learnable specialized expert tokens, $\mathbf{Q} \in \mathbb{R}^{N_a\times d_c}$, where N_a denotes the number of abnormalities. First, attention scores are computed by performing a scaled dot-product operation between the specialized expert tokens $\mathbf{Q}$ and the refined common expert representations $\mathbf{E}'$, as shown in (6). The softmax function normalizes these attention weights across all common expert tokens, ensuring that the model focuses on the most relevant features. To improve numerical stability, the computed scores are scaled by a factor of $\frac{1}{\sqrt{d_c}}$. Using these attention scores, the common expert token representations $\mathbf{E}'$ are aggregated through weighted summation, resulting in the final pooled representations $\mathbf{P}$, as defined in (7).

$$\mathbf{A} = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{E}'^{\top}}{\sqrt{d_c}}\right), \tag{6}$$

$$\mathbf{P} = \mathbf{A}\mathbf{E}', \tag{7}$$

Classifier: The pooled abnormality-specific representation $\mathbf{P}$ is passed through independent classification heads, each implemented as a multi-layer perceptron (MLP). These

classification heads predict the presence or absence of each abnormality and incorporate an uncertainty label, facilitating multi-class, multi-label classification.

For each abnormality j, the pooled representation $\mathbf{P}_j$ is transformed into logits using an MLP as defined in (8).

$$\mathbf{Z}_j = \mathrm{MLP}_j(\mathbf{P}_j), \quad j = 1, 2, \ldots, N_a \tag{8}$$

where $\mathbf{Z}_j \in \mathbb{R}^{B\times 3}$ represents the logits corresponding to the three possible labels: positive, negative, and uncertain.

The logits are then converted into probability distributions using the softmax activation function as defined in (9)

$$\hat{\mathbf{Y}}_j = \mathrm{softmax}(\mathbf{Z}_j) \tag{9}$$

where $\hat{\mathbf{Y}}_j \in \mathbb{R}^{B\times 3}$ denotes the predicted probability distribution for the jth abnormality.

The following section details the objective functions utilized for abnormality-specific feature extraction and classification during the pre-training phase.

Loss Functions:

d: CLASSIFICATION LOSS (CLS)

To ensure accurate abnormality predictions, the classification loss is computed using a class-weighted cross-entropy function, which accounts for class imbalance by assigning different weights to each label. Given the predicted probability distribution $\hat{\mathbf{Y}}_j$ obtained from the softmax output in Equation (9), the classification loss for each abnormality j is defined as given in (10):

$$L_j = \frac{1}{B}\sum_{i=1}^{B} -w_{y_{i,j}} \cdot Y_{i,j} \log(\hat{\mathbf{Y}}_{i,j}), \tag{10}$$

where B is the batch size, $Y_{i,j} \in \{0, 1, 2\}$ represents the ground truth label for sample i and abnormality j, $\hat{\mathbf{Y}}_{i,j} \in \mathbb{R}^3$ is the predicted probability distribution over the three possible labels: positive, negative, and uncertain, and $w_{y_{i,j}}$ is the class weight associated with the ground truth label.

The final classification loss is computed by averaging across all abnormalities as given in (11):

$$L_{\mathrm{cls}} = \frac{1}{N_a}\sum_{j=1}^{N_a} L_j. \tag{11}$$

The class-weighted approach ensures that underrepresented abnormality classes contribute proportionally to the overall loss, preventing bias toward majority classes.

e: ORTHOGONALITY LOSS (OL)

To enhance the discriminative capacity of common expert token representations and reduce redundancy among them, we incorporate an orthogonality constraint. This constraint enforces *decorrelation* across the refined common expert tokens by penalizing their pairwise similarity, thereby ensuring that each token captures a distinct set of features during the cross-attention process. Let $\mathbf{E}' \in \mathbb{R}^{N\times D}$ represent the matrix of refined common expert token embeddings, where N is the number of tokens and D is the embedding dimension. We compute the *Gram matrix* $\mathbf{S} \in \mathbb{R}^{N\times N}$ as defined in (12):

$$\mathbf{S} = \mathbf{E}' \cdot \mathbf{E}'^{\top}, \tag{12}$$

The matrix $\mathbf{S}$ contains all pairwise inner products among expert tokens. Ideally, $\mathbf{S}$ should approximate an identity matrix, indicating orthogonality. To encourage this, we minimize the squared off-diagonal elements across the batch as defined in (13):

$$L_{\mathrm{orth}} = \frac{1}{B}\sum_{b=1}^{B}\sum_{i\neq j}(\mathbf{S}_{b,i,j})^2, \tag{13}$$

where B is the batch size and $\mathbf{S}_{b,i,j}$ denotes the similarity between expert tokens i and j for sample b. This regularization reduces feature overlap and enhances representation disentanglement, which has been shown to benefit both classification and interpretability [58], [59].

f: SPARSITY LOSS (SL)

To encourage each common expert token to focus on the most informative regions of the input image, we introduce a sparsity constraint on the attention maps. This regularization promotes selective activation by penalizing uniformly distributed attention, thereby compelling the model to concentrate on the most salient features. Similar entropy-based regularization has been applied in cross-modal similarity contexts, as in TIER [60], where entropy minimization encourages focused patch-text associations. Inspired by this, our sparsity loss applies the concept directly to intra-modal attention scores within the MHCAC module.

Let $\mathbf{A}_l \in \mathbb{R}^{N\times M}$ denote the attention map at layer l, where N is the number of tokens and M is the number of spatial positions. The sparsity loss is defined as given in (14), (15):

$$L_{\mathrm{sparsity},l} = -\frac{1}{|\mathbf{A}_l|}\sum_{i,j}\mathbf{A}_{l,i,j}\log(\mathbf{A}_{l,i,j} + \epsilon), \tag{14}$$

$$L_{\mathrm{sparsity}} = \frac{1}{L}\sum_{l=1}^{L} w_l L_{\mathrm{sparsity},l} \tag{15}$$

where w_l is the layer-wise weighting coefficient, and ϵ is a small constant to prevent numerical instability. This entropy-based formulation minimizes uncertainty in the attention distribution, encouraging the model to place a sharper focus on key regions relevant to abnormality classification.

g: CONTRASTIVE LOSS (CL)

To enhance the discriminability of abnormality-specific representations, we adopt a supervised contrastive loss inspired by [57]. This loss encourages better separation between positive, negative, and uncertain samples in the learned feature space. Unlike traditional contrastive learning, which focuses solely on binary class separation, our formulation

explicitly accounts for uncertain samples to better reflect clinical uncertainty.

Let $\mathbf{P} \in \mathbb{R}^{N \times d}$ represent the pooled abnormality-specific representation after the attention pooling layer. We compute the pairwise similarity matrix as defined in (16):

$$\mathbf{S}_{\text{pairwise}} = \mathbf{P} \cdot \mathbf{P}^{\top}, \tag{16}$$

For each abnormality class, the contrastive loss consists of two parts. The first term promotes separation between positive and negative class samples as defined in (17):

$$L_{\text{pos-neg}} = \frac{1}{N_p N_n} \sum_{i \in \text{pos}, j \in \text{neg}} \text{ReLU}(m - (1 - \mathbf{S}_{i,j})), \tag{17}$$

where N_p and N_n are the numbers of positive and negative samples, respectively, and m is a predefined margin.

The second term handles uncertain cases by ensuring their representations remain equidistant from both positive and negative class centroids, as given in (18):

$$L_{\text{unc}} = \frac{1}{N_u} \sum_{i \in \text{unc}} \left| \text{Mean}(\mathbf{S}_{i,\text{pos}}) - \text{Mean}(\mathbf{S}_{i,\text{neg}}) \right|, \tag{18}$$

where N_u is the number of uncertain samples.

Finally, the overall contrastive loss is computed by averaging across all N_a abnormality classes as given in (19):

$$L_{\text{contrastive}} = \frac{1}{N_a} \sum_{a=1}^{N_a} \left(L_{\text{pos-neg},a} + L_{\text{unc},a} \right). \tag{19}$$

h: TOTAL LOSS

The final optimization objective combines all loss components as formulated in (20):

$$L = L_{\text{cls}} + \lambda_{\text{orth}} L_{\text{orth}} + \lambda_{\text{sparsity}} L_{\text{sparsity}} + \lambda_{\text{contrastive}} L_{\text{contrastive}}. \tag{20}$$

E. MODEL PRE-TRAINING AND FINE TUNING

The pre-training of the META-CXR model is conducted in two distinct phases, as illustrated in Figure 3. The first phase focuses on *Representation Learning*, followed by the second phase dedicated to *Generative Learning*. Both phases are trained using the MIMIC-CXR-JPG [49] training split.

1) REPRESENTATION LEARNING PHASE

During the representation learning phase, ITC, ITG, ITM, and MHCAC module-specific CLS loss and other auxiliary losses are used as objective functions where both the MHCAC module and META-Former are kept unfrozen while vision encoders and Language Model(LM) are kept frozen. For better medical domain-specific feature extraction, we utilize the SWIN transformer-based MedClipViT [61] and Bio-ViL-T image encoder [62], which uses ResNet50 as the backbone feature extractor, as well as the ViT-based PubMedCLIP [63] vision encoder, all of which have been fine-tuned using radiology data. For all the vision encoders, we utilized the pre-processing techniques specific to each encoder and we set the visual feature dimension after linear projection to 1408 for concatenation, which is ideal for CNN, ViT, and SWIN transformer-based features. The dimension of the ViT features is 50 × 1048, and for the SWIN, the feature dimension is 50 × 1408, including the feature vector aggregated through global pooling. For the ResNet50-based model, we utilize the layer before the global average pooling to extract the spatial feature map and flatten it out to produce patch-wise embeddings resulting in a 196 × 1408 output after the linear projection. As a regularization technique, we randomly mask 10% of the features from each encoder.

We implemented META-Former by modifying the original Q-Former architecture from BLIP-2 [10] to accept concatenated visual features from multiple vision encoders, while keeping all other architectural components unchanged. The modified META-Former is initialized using the pretrained Q-Former weights to preserve its learned multimodal alignment capabilities. In the MHCAC module, we introduce a set of shared common expert tokens along with 14 abnormality-specific learnable expert query tokens, each initialized randomly. All expert tokens are set to a hidden dimension of 768.

During the initial training phase, we freeze the MHCAC module, allowing the META-Former-specific query tokens to focus on extracting multimodal knowledge from the combined visual features and text reports. This stage optimizes only the ITC, ITG, and ITM objectives. Subsequently, we progressively unfreeze the MHCAC module while gradually freezing the META-Former parameters, enabling fine-tuning of the classification-specific expert tokens, multi-head cross-attention layers, and classification heads. During this stage, we employ a warm-up learning rate between 5e-5 and 2e-4, apply L2 regularization, and set a dropout rate of 0.2. A cosine annealing learning rate schedule is then applied, decaying from 2e-4 to 1e-5 over five epochs.

Once initial convergence on the classification task is achieved, we also unfreeze the META-Former parameters, allowing the projection heads of the visual encoders to be fine-tuned jointly for both classification and report generation tasks. This phase leverages a weighted loss function to balance the learning objectives effectively.

2) GENERATIVE LEARNING PHASE

In the generative learning phase, META-Former output is connected to a Language Model (LM) to leverage the LLM's generative language capabilities. The META-Former query outputs are linearly projected through a fully connected (FC) layer into the same dimension as the LM's text embedding. We then use a prompt construction method where the query token embeddings, abnormality findings from the MHCAC module, and natural language instructions are combined to create a single prompt for the LLM.

3) PROMPT

Image Features: `<image queries>` *Positive Abnormalities*: `<positive results>` *Negative Abnormalities*:

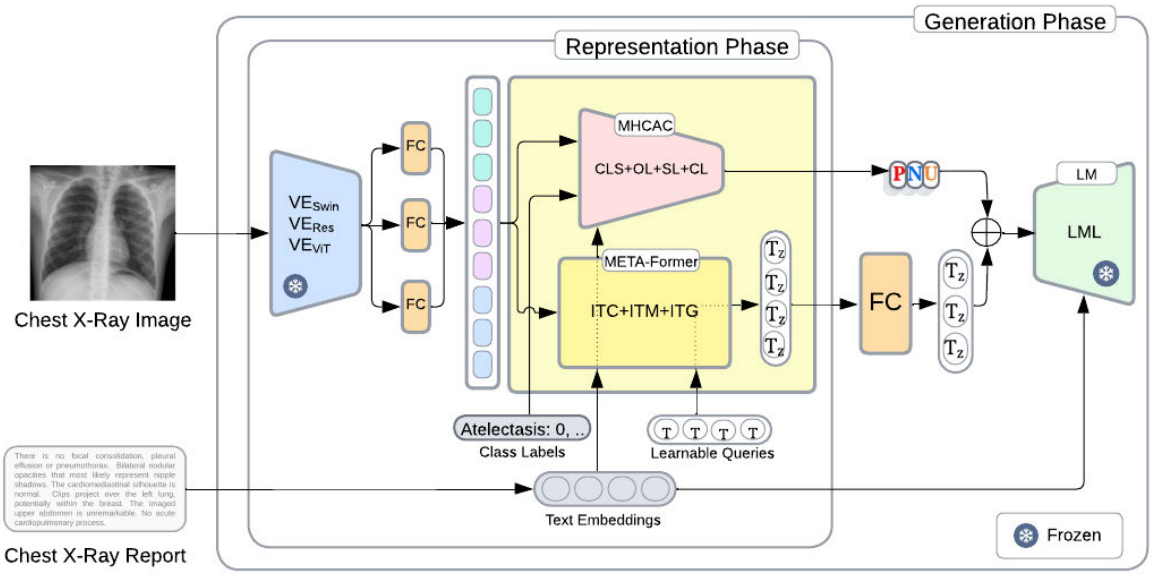


FIGURE 3. Learning phases of META-CXR. The loss functions associated with each component—MHCAC, META-Former, and the Language Model (LM)—are indicated within their respective modules.

`<negative results>` *Uncertain Abnormalities*: `<uncertain results>` *Generate the radiology report for the given chest X-ray image*

We only consider abnormalities with a confidence score higher than the relevant threshold for each abnormality as depicted in figure 11 when feeding the information to the LM, ensuring that the abnormality supervision is precise and effective. The LLM then generates the corresponding radiology report by taking into account the image queries and abnormal findings.

To enhance medical report generation, we integrated the Vicuna 7B LLM with LoRA adapters following the approach in [17], which had been fine-tuned on the MIMIC-CXR-JPG training dataset. To further align the model with our specific prompting strategy, we performed instruction tuning using a structured prompt format that explicitly incorporates the "uncertain" class. This tuning phase optimized a language modeling loss (LML) as the objective function, where only the projection head from the META-Former output and the LoRA layers of the LLM were updated. During this phase, we set the LoRA rank to 8 and the LoRA alpha to 16.

We conducted pre-training for 20,000 steps during the representation learning phase and 5,000 steps during the generative learning phase. The AdamW optimizer was used throughout all training stages. All training and fine-tuning experiments were performed on a single NVIDIA RTX 4070 (16 GB) GPU. Our implementation is based on PyTorch and the Hugging Face Transformers library. To support transparency and reproducibility, we provide the complete source code—including model architecture, training routines, and inference pipelines—at our GitHub repository [64].

Algorithm 1 Inference Process of META-CXR

Require: *Chest radiograph* $\mathbf{X}$
Ensure: *Generated radiology report* $\mathbf{R}$, *Abnormality Findings* $\mathbf{F}$, *Attention maps* $\mathbf{A}$

1: **Step 1: Visual Feature Extraction**
2: $\mathbf{V} \leftarrow$ *Extract enriched and concatenated visual features from* $\mathbf{X}$ *using CNN, ViT, Swin Transformer*
3: **Step 2: Extract Visual Features for Report Generation via META-Former**
4: $\mathbf{Q}_M \leftarrow$ *Cross-attend visual features* $\mathbf{V}$ *with learned query embeddings via META-Former*
5: **Step 3: Abnormality Classification via MHCAC**
6: $\mathbf{E}' \leftarrow$ *Extract generalized abnormality features from* $\mathbf{V}$ *using common expert tokens* $\mathbf{E}_{\text{common}}$
7: $\mathbf{P} \leftarrow$ *Extract abnormality-specific pooled representation from* $\mathbf{E}'$ *using specialized expert tokens* $\mathbf{Q}$
8: $\mathbf{F} \leftarrow$ *Classify abnormalities using pooled representation and confidence thresholds*
9: **Step 4: Soft Prompt Construction**
10: $\mathbf{S} \leftarrow$ *Integrate query embeddings* $\mathbf{Q}_M$ *with abnormality findings* $\mathbf{F}$
11: **Step 5: Radiology Report Generation via LLM**
12: $\mathbf{R} \leftarrow$ *Generate comprehensive radiology report by providing soft prompt* $\mathbf{S}$ *to fine-tuned LLM*
13: **Step 6: Attention Map Extraction**
14: $\mathbf{A} \leftarrow$ *Retrieve attention maps from MHCAC module for interpretability*
15: **return** $\mathbf{R}, \mathbf{F}, \mathbf{A}$

F. MODEL INFERENCE

1) INFERENCE PIPELINE

The inference process of META-CXR follows a structured pipeline, as outlined in Algorithm 1, to generate clinically relevant radiology reports from chest radiographs.

2) SOLUTION DEPLOYMENT

To support reproducibility and ease of use, we provide a complete deployment pipeline for META-CXR, including a user-friendly interface, public codebase, and Docker-based runtime environment. A Gradio-powered web UI has been developed to allow users to upload chest X-ray images and interactively view the generated radiology reports and classification results. An example of this interface is shown in Figure 4.

The full implementation is publicly available via GitHub [64], which includes pretrained model weights, configuration files, and inference scripts. To simplify setup and eliminate environment mismatches, we also provide a Docker container, preconfigured with all necessary dependencies for running META-CXR. Users only need to specify the path to the model weights and the configuration YAML file to initiate the inference pipeline.

We recommend deploying the Docker container on a machine with at least 16 GB of GPU VRAM to ensure stable performance during both classification and report generation tasks.

IV. EXPERIMENTAL RESULTS

A. EXPERIMENTAL SETUP

For evaluation, we focus on two core tasks—radiology report generation and abnormality classification—conducted across multiple publicly available chest X-ray datasets. The MIMIC-CXR-JPG test set [49], consisting of 3,269 radiographs with paired radiology reports and abnormality labels, serves as the primary evaluation dataset. To assess cross-dataset generalizability in report generation, we additionally include the IU-Xray test set [28]. We adopt the conventional dataset partitioning strategy from prior work (e.g., MeTransformer [37]) to ensure fair and consistent comparison. For evaluating abnormality classification beyond the training distribution, we use the CheXpert validation set [29], comprising 200 manually labeled chest radiographs. Evaluation metrics used for each task are selected to balance natural language fidelity and clinical relevance, and are explained in detail in subsequent subsections. Training configurations and hyperparameters are described separately in Section III-E: Model Pre-Training and Fine Tuning.

B. QUANTITATIVE RESULTS

1) REPORT GENERATION RESULTS

To comprehensively assess the quality and clinical utility of generated radiology reports, we employ a combination of Natural Language Generation (NLG) metrics and radiology-specific evaluation metrics. These collectively measure linguistic fluency, semantic similarity, factual correctness, and clinical efficacy.

Natural Language Generation (NLG) Metrics

- **BLEU (1–4)** [65]: The Bilingual Evaluation Understudy (BLEU) score calculates n-gram precision between generated and reference texts. BLEU-1 to BLEU-4 assess unigram to 4-gram overlaps, respectively, capturing increasing levels of textual coherence.
- **ROUGE-L** [66]: Recall-Oriented Understudy for Gisting Evaluation (ROUGE-L) evaluates the longest common subsequence between generated and reference texts, focusing on recall and fluency.
- **METEOR** [67]: The Metric for Evaluation of Translation with Explicit Ordering (METEOR) computes a harmonic mean of unigram precision and recall, incorporating synonym matching and stemming to better reflect linguistic meaning.
- **CIDEr** [68]: Consensus-based Image Description Evaluation (CIDEr) measures the similarity of generated text to a set of references using TF-IDF weighted n-grams, capturing consensus-level generation quality.

Radiology-Specific Metrics

- **BERTScore** [69]: BERTScore uses contextual embeddings from a pre-trained BERT model to compute semantic similarity between generated and reference reports. It better captures meaning than surface-level metrics.
- **RadGraph F1** [70]: This metric quantifies the overlap of clinical entities and relations between generated and reference reports, based on structured graph extraction. It reflects the factual accuracy of clinical content.
- **RadCliQ** (↓) [71]: A composite clinical quality score derived from radiologist feedback and NLP analysis. It penalizes clinically incorrect or irrelevant statements. Lower scores indicate higher clinical coherence and safety.
- **Clinical Efficacy (CE) Metrics** [71]: Following the CheXpert [29] labeling protocol, we assess how well generated reports convey the presence of 14 common radiological abnormalities. We report:
  - **Precision**: Proportion of correctly predicted positive findings out of all positive predictions.
  - **Recall**: Proportion of true positive findings identified out of all actual positives.
  - **Macro F1-Score**: Harmonic mean of precision and recall, averaged across all abnormality classes to ensure balanced evaluation.

We compare META-CXR with state-of-the-art (SOTA) vision-language model (VLM)-based radiology report generation (R2G) methods, including R2GenGPT [72], KARGEN [39], Flamingo-CXR [35], Med-PaLM M [73], MAIRA-2 [33], and Ra-Dialog [17]. Additionally, we evaluate its performance against other SOTA medical report generation models, such as METransformer [37] and Kiut [45].

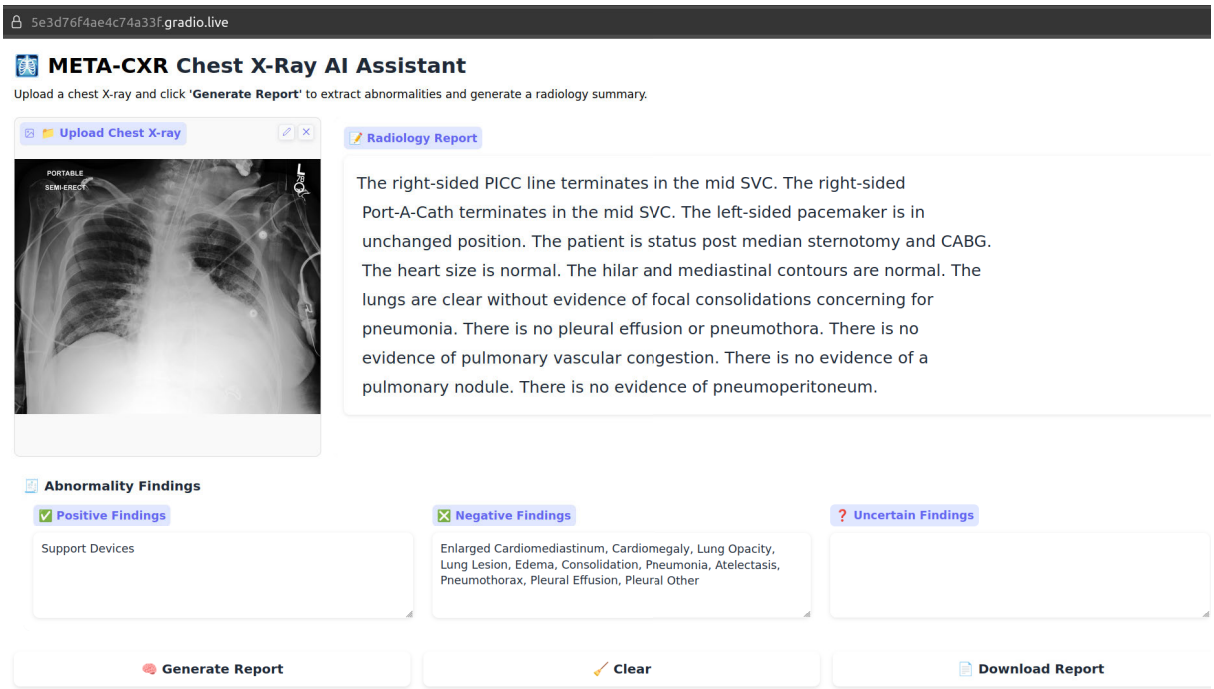


FIGURE 4. Gradio-based web interface for running META-CXR locally. Users can upload chest X-ray images and receive detailed radiology reports and classification results.

TABLE 2. Standard NLP metrics on MIMIC-CXR and IU-XRay test sets.

Dataset	Method	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE-L	CIDEr
MIMIC-CXR	METransformer [37]	0.379	0.245	0.171	0.124	0.152	0.291	**0.362**
	KiUT [45]	0.362	0.233	0.162	0.113	0.160	0.285	–
	R2GenGPT [72]	0.401	0.264	0.185	0.134	0.160	0.297	0.269
	Med-PaLM 12B [73]	0.351	0.238	0.162	0.104	–	0.262	0.234
	Flamingo-CXR [35]	0.347	0.230	0.158	0.101	–	0.297	0.138
	RaDialog-RG [17]	0.360	0.240	0.168	0.095	0.140	0.271	–
	KARGEN [39]	0.417	**0.278**	**0.192**	0.140	0.165	0.305	0.289
	MAIRA-2 [33]	**0.460**	–	–	**0.231**	**–**	**0.384**	**–**
	META-CXR (Ours)	0.390	0.255	0.175	0.102	**0.173**	0.280	0.291
IU-XRay	METransformer [37]	0.483	0.322	0.228	0.172	0.192	0.380	0.435
	KiUT [45]	**0.525**	**0.360**	**0.251**	**0.185**	**0.242**	**0.409**	**–**
	R2GenGPT [72]	0.488	0.316	0.228	0.173	0.211	0.377	0.438
	KARGEN [39]	0.490	0.323	0.232	0.180	0.218	0.385	**0.491**
	MAIRA-2$^{(OOD)}$ [33]	–	–	–	0.117	–	0.274	–
	META-CXR (Ours) $^{(OOD)}$	0.408	0.221	0.198	0.121	0.154	0.287	0.413
	META-CXR (Ours)	0.478	0.293	0.208	0.160	0.189	0.319	0.467

a: EVALUATION ON MIMIC-CXR

META-CXR demonstrates balanced performance across both standard and clinical-specific metrics on the MIMIC-CXR test set. As shown in Table 2, although KARGEN and MAIRA-2 achieve the highest BLEU-1 and BLEU-4 scores, META-CXR achieves the best METEOR score (0.173), indicating stronger semantic alignment. Its BLEU-4 score (0.102) is modest, likely due to more diverse linguistic expression that prioritizes semantic fidelity over strict lexical overlap. META-CXR also reports competitive ROUGE-L (0.280) and CIDEr (0.291) scores, with CIDEr outperforming all other VLMs, reflecting stronger consensus-based content relevance.

As shown in Table 3, META-CXR also performs well on clinical-specific metrics. It achieves the highest BERTScore (0.426), surpassing KARGEN (0.421) and R2GenGPT (0.415), confirming the model's ability to generate semantically faithful and medically coherent reports. It also reports a RadGraph F1 score of 0.224, outperforming both Flamingo-CXR and KARGEN, and demonstrating better

TABLE 3. Clinical and domain-specific metrics on MIMIC-CXR test sets.

Dataset	Method	NLP-Derived Clinical Metrics			Clinical Efficacy Metrics		
		BERTScore	RadGraph F1	RadCliQ(↓)	Precision	Recall	Macro F1
MIMIC-CXR	METransformer [37]	–	–	–	0.364	0.309	0.311
	KiUT [45]	–	–	–	0.371	0.318	0.321
	R2GenGPT [72]	0.415	0.187	1.207	0.392	0.387	0.389
	Med-PaLM 12B [73]	–	0.252	–	–	–	0.373
	Flamingo-CXR [35]	–	0.205	–	–	–	–
	RaDialog-RG [17]	0.400	–	–	–	–	0.394
	KARGEN [39]	0.421	0.203	**1.165**	–	–	–
	MAIRA-2 [33]	-	**0.346**	2.640	–	–	0.416
	META-CXR (Ours)	**0.426**	0.224	1.211	**0.411**	**0.465**	**0.428**

encoding of clinical entities and relationships. Although META-CXR yields a slightly higher (worse) RadCliQ score (1.211) compared to KARGEN (1.165), it achieves the best Clinical Efficacy F1-score (0.428), reflecting more accurate alignment between abnormality findings and the generated text, driven by its classification-aware report generation approach.

b: EVALUATION ON IU-XRAY

As presented in Table 2, META-CXR is evaluated on the IU-XRay test dataset under two configurations: out-of-distribution (OOD) and instruction-tuned. In the OOD setting, META-CXR outperforms MAIRA-2 (OOD) with higher BLEU-4 (0.121 vs. 0.117) and ROUGE-L (0.287 vs. 0.274), demonstrating stronger cross-domain generalization, which can be attributed to its abnormality-guided architecture and multi-encoder feature extraction.

Following instruction tuning on IU-XRay data, META-CXR shows consistent improvements across all standard metrics: BLEU-4 increases to 0.160, METEOR to 0.189, ROUGE-L to 0.319, and CIDEr to 0.467. These gains reflect the model's ability to adapt to dataset-specific linguistic structures and reporting conventions. However, compared to fully fine-tuned models such as KiUT, METransformer, and KARGEN, META-CXR still lags slightly in lexical alignment and phrasing quality. This suggests that, while instruction tuning is effective, further gains are likely attainable through full end-to-end fine-tuning on the IU-XRay dataset.

2) ABNORMALITY CLASSIFICATION RESULTS

We evaluated the performance of our multi-label, multi-class classifier using four key metrics: Precision, Recall, F1-score, and Area Under the Receiver Operating Characteristic Curve (AUC).

In the context of medical image classification, **Recall** (or sensitivity) quantifies the proportion of actual positive cases that are correctly identified by the model, i.e., true positives (TP) over all ground truth positives. **Precision** measures the proportion of correctly predicted positive cases among all positive predictions, reflecting the model's reliability in identifying true pathology when it predicts a condition. The **F1-score**, as the harmonic mean of precision and recall, provides a balanced measure of performance, especially in imbalanced datasets common in clinical settings.

Additionally, we report the **AUC**, which reflects the model's ability to distinguish between classes across all classification thresholds. A higher AUC indicates better discriminatory power, particularly important for evaluating model performance in the presence of uncertain or ambiguous findings in chest radiographs.

a: EVALUATION ON MIMIC-CXR DATASET

We evaluate the classification performance of META-CXR using the MIMIC-CXR-JPG [49] test dataset to assess its effectiveness in multi-class abnormality classification. To the best of our knowledge, there is no prior literature that provides a direct comparison for uncertainty-aware abnormality classification results on the MIMIC-CXR test set.

META-CXR achieves strong classification performance, with an average precision of 0.87, recall of 0.78, and F1-score of 0.73, demonstrating its ability to accurately identify and distinguish between different abnormality classes.

The AUC-ROC curves as depicted in Figure 5 highlight the robustness of META-CXR in multi-class classification, particularly with the explicit incorporation of an uncertainty class, which enhances the model's ability to handle real-world clinical ambiguity. Notably, tasks such as "Support Devices" achieve an exceptionally high AUC of 0.92 (positive) and 0.84 (negative), demonstrating the model's strong capacity for distinguishing between these cases. Similarly, "Pleural Effusion" achieves high AUC values of 0.85 (negative) and 0.74 (positive) and "Edema" (AUC 0.78 negative), underscoring its effectiveness in detecting complex diagnostic categories.

b: CROSS-DOMAIN PERFORMANCE EVALUATION ON CHEXPERT DATASET

Cross-domain classification is essential in medical AI, allowing models to generalize to new datasets without retraining. We evaluate the cross-domain classification performance

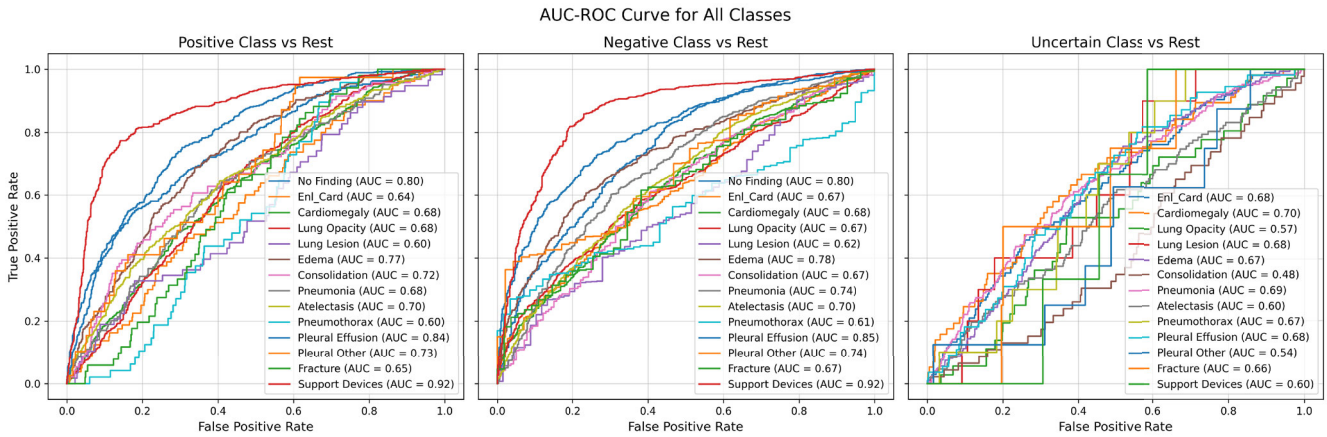


FIGURE 5. AUC-ROC Curves on MIMIC-CXR test set for all 14 pathologies including No Finding.

of META-CXR on the CheXpert [29] validation set, and compare it with two zero-shot baselines: CheXZero [74] and ELIXR-C [16]. Results are reported for five key pathologies using AUC and F1-score metrics.

CheXZero and ELIXR-C are zero-shot models that use CLIP-like vision-language contrastive learning. They embed image and text pairs into a shared space and infer labels by computing the similarity between an image and handcrafted prompts.

In contrast, META-CXR is trained specifically for multi-class, multi-label abnormality classification. Its *Multi-Head Cross-Attention Classification (MHCAC)* module includes explicit modeling of `positive`, `negative`, and `uncertain` labels for each finding, aligning with real-world radiology reporting. While not zero-shot, it supports robust cross-domain classification without data-specific fine-tuning.

For fair evaluation, we convert the multi-class predictions into binary form as given in (21):

$$p_{\text{final}} = \frac{p_1}{p_0 + p_1} \tag{21}$$

This retains the benefit of uncertainty modeling during training but aligns with the binary outputs expected in the CheXpert evaluation.

As shown in Table 4, META-CXR achieves a higher mean F1-score (0.699) than CheXZero (0.606), particularly outperforming on complex findings like Consolidation (0.792 vs. 0.333) and Edema (0.721 vs. 0.602). Although CheXZero leads in AUC, META-CXR offers a better balance between precision and recall, benefiting from structured abnormality-specific heads and uncertainty awareness.

C. QUALITATIVE RESULTS

1) REPORT GENERATION EFFECTIVENESS

Figure 6 presents a comparison between the ground truth and the report generated by META-CXR on the MIMIC-CXR-JPG [49] test dataset, along with the corresponding classification outputs. The same color is used to highlight abnormality-specific text within the report. Additionally, the impact of including(PNU) vs excluding(PN) the uncertainty class in prompt construction is examined.

Additionally, the attention maps of 8 common expert tokens are stacked and displayed alongside the generated report. These attention maps highlight the regions of the radiograph attended to by the relevant expert tokens, providing valuable insights into the model's interpretability and decision-making process.

2) MODEL EXPLAINABILITY AND ATTENTION MAP

To enhance model interpretability, we extract and visualize attention maps from the cross-attention layers of the MHCAC module, focusing on the 8 common expert tokens. These expert tokens interact with the fused visual features obtained from multiple encoders, including ResNet50, ViT, and Swin Transformer. The cross-attention mechanism computes relevance scores between each expert token (as a query) and the aggregated visual tokens (as keys and values), followed by a softmax operation to yield normalized attention distributions.

As illustrated in Figure 7, we generate heatmaps by overlaying attention scores of each expert token onto the ResNet50 feature grid of a sample chest X-ray from the MIMIC-CXR test set. These maps highlight spatial regions deemed most relevant by each expert token, revealing how the model attends to specific anatomical areas. Since the MHCAC module is jointly optimized for both classification and report generation, these attention maps serve as a transparent window into the model's internal decision-making. This visualization not only aids in understanding the reasoning behind each predicted abnormality but also builds clinician trust and facilitates downstream clinical validation.

D. ABLATION STUDY

In this section, we conduct ablation studies on the MIMIC-CXR test dataset to investigate the contribution of each component in META-CXR.

TABLE 4. Cross-domain classification performance of the MHCAC module on five pathologies in the CheXpert dataset, compared with SOTA methods.

Metric	Model	Mean	Atelectasis	Cardiomegaly	Consolidation	Edema	Pleural Effusion
AUC	CheXZero [74]	**0.889**	**0.816**	**0.906**	**0.892**	**0.897**	**0.932**
	ELIXR-C [16]	0.851	0.754	0.891	0.875	0.880	0.930
	META-CXR (Ours)	0.824	0.792	0.842	0.790	0.820	0.878
F1-Score	CheXZero [74]	0.606	**0.646**	**0.743**	0.333	0.602	**0.704**
	ELIXR-C [16]	–	–	–	–	–	–
	META-CXR (Ours)	**0.699**	0.623	0.674	**0.792**	**0.721**	0.689

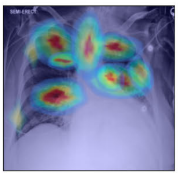
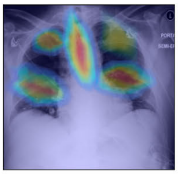
Radiograph	Ground Truth	META-CXR (PN)	META-CXR (PNU)	Stacked Attention Map
	Abnormalities: **Positive**: Lung opacity, Edema, support device **Uncertain**: None **Negative**: Rest **Report**: AP radiograph of the chest was reviewed in comparison to ___. The ET tube tip is . cm above the carina. NG tube tip is in the stomach. There is left retrocardiac opacity, unchanged since the prior study. Minimal interstitial pulmonary edema is unchanged. No interval development of pleural effusion or pneumothora is seen.	**Predicted Abnormalities:** **Positive**: Lung opacity, Edema, support device **Uncertain**: None **Negative**: Rest **Report**: Retrocardiac opacity in the left lung. Minimal interstitial pulmonary edema is present without interval change. Endotracheal tube tip is positioned. Nasogastric tube is appropriately located in the stomach. No pleural effusion or pneumothorax. Mediastinum and bony structures are normal.	**Predicted Abnormalities:** **Positive**: Lung opacity, Edema, support device **Uncertain**: None **Negative**: Rest **Report:** Retrocardiac opacity in the left lung. Minimal interstitial pulmonary edema is present without interval change. Endotracheal tube tip is positioned. Nasogastric tube is appropriately located in the stomach. No pleural effusion or pneumothorax. Mediastinum and bony structures are normal.	
	Abnormalities: **Positive**: None **Uncertain:** Cardiomegaly, Enlarged Cardiomediastinum **Negative:** Rest **Report:** AP radiograph of the chest was reviewed in comparison to CT torso from the same day obtained earlier. There is prominence of the right paratracheal stripe consistent with known enlarged lymph node. Heart size and mediastinum **are otherwise unchanged** in the short interim. Calcified granuloma in the right lower lobe is redemonstrated. No new abnormalities within the lungs seen.	**Predicted Abnormalities:** **Positive:** None **Uncertain:** None **Negative:** Rest **Report:** Right paratracheal stripe appears prominent, consistent with an enlarged lymph node. A calcified granuloma is present in the right lower lobe without change. The cardiomediastinal contour remains stable. Heart size is normal. No acute pulmonary findings. No pleural effusion or pneumothorax.	**Predicted Abnormalities:** **Positive:** None **Uncertain:** Cardiomegaly, Enlarged Cardiomediastinum **Negative:** Rest **Report:** Prominence of the right paratracheal stripe, consistent with an enlarged lymph node. A calcified granuloma is noted in the right lower lobe and remains unchanged. The cardiomediastinal contour is stable. Heart size is within normal limits, though cardiomegaly remains uncertain. No acute pulmonary abnormalities. No pleural effusion or pneumothorax.	

FIGURE 6. Report generation comparison with ground truth on the MIMIC-CXR test dataset. 'PN' considers only positive and negative classes, whereas 'PNU' includes all three classes (positive, negative, and uncertain) in prompt construction.

1) EFFECT OF COMBINED VISION ENCODER

To assess the contribution of each vision encoder in the proposed multi-encoder architecture, we conduct a two-stage ablation study focusing on (a) abnormality classification and (b) report generation. Additionally, we perform qualitative analysis using attention maps to interpret how expert tokens interact with features from distinct encoder families.

a: EFFECT ON ABNORMALITY CLASSIFICATION

We evaluate the classification performance by selectively activating one encoder at a time during inference within the MHCAC module while keeping the common expert tokens unchanged. This isolates the contribution of each encoder's feature representation to the final abnormality classification. We only use 5 common pathologies for this evaluation.

Table 5 summarizes the mean F1 scores for each encoder individually, as well as for the full multi-encoder configuration. The results show that while each encoder is capable of capturing meaningful features, the combination significantly improves classification performance. This demonstrates the complementary nature of the inductive biases encoded by CNNs, ViTs, and Swin Transformers.

b: EFFECT ON REPORT GENERATION

To isolate the impact of encoder diversity on report generation, we fix the classification outputs and evaluate the

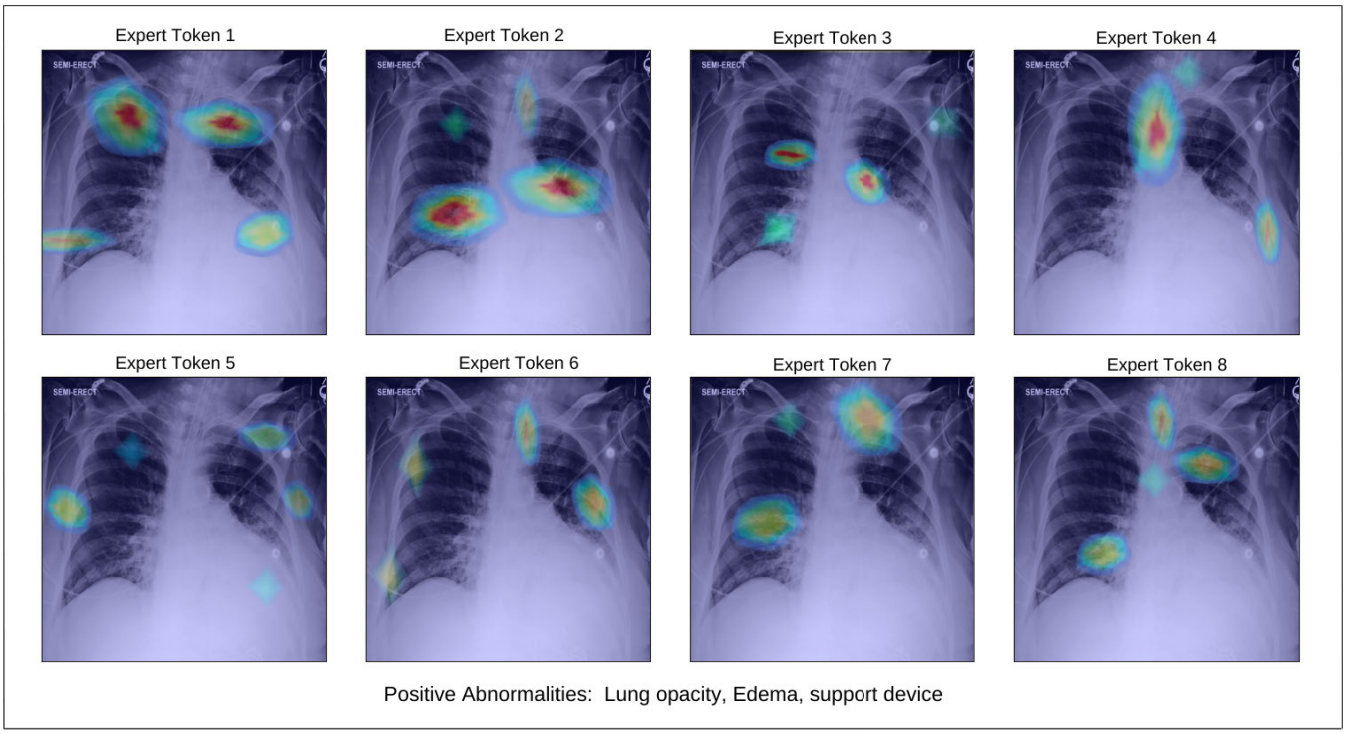


FIGURE 7. Attention maps of 8 common expert tokens from the MHCAC module over ResNet50 features for a MIMIC-CXR test set sample, highlighting token-specific focus regions.

TABLE 5. Mean F1 score across 5 common abnormalities for abnormality classification under different encoder configurations on the MIMIC-CXR test set.

Vision Encoders			Mean F1 Score
RN50	ViT	Swin	
✓	–	–	0.602
–	✓	–	0.473
–	–	✓	0.467
✓	✓	–	0.631
✓	–	✓	0.682
✓	✓	✓	**0.701**

TABLE 6. Report Generation under different encoder configurations on the MIMIC-CXR test set.

Vision Encoders			BERTScore
RN50	ViT	Swin	
✓	–	–	0.312
–	✓	–	0.289
–	–	✓	0.267
✓	✓	–	0.401
✓	–	✓	0.394
✓	✓	✓	**0.426**

META-Former module by using image features from only one encoder at a time. That is, during inference, only one encoder's features are passed through to the generation head while all other components—MHCAC module, soft prompt tokens, and LLM—remain unchanged.

We use the BERTScore for this study since this metric evaluates the semantic relevance and clinical accuracy of the generated reports by comparing them to reference reports. The results in Table 6 indicate that combining CNN, ViT, and Swin encoders yields complementary visual features, enhancing the quality of generated radiology reports.

c: QUALITATIVE VALIDATION VIA ATTENTION MAPS

To provide interpretability for the encoder contributions, we visualize attention heatmaps from the MHCAC module for a fixed set of 4 common expert tokens across all three encoders. Figure 8 presents these side-by-side comparisons on the same chest X-ray image.

Interestingly, we observe that while all three encoders—ResNet (CNN), ViT, and Swin—attend to the same clinically relevant anatomical regions for a given expert token, the **intensity of attention differ across encoders**. This supports the hypothesis that expert tokens are learning to aggregate complementary information from different encoder types.

Together, these quantitative metrics and qualitative attention maps demonstrate that our multi-encoder architecture enhances both classification and generation tasks, leveraging the inductive biases of distinct encoder families to improve robustness and interpretability.

2) EFFECT OF LOSS FUNCTIONS IN MHCAC

To evaluate the contribution of each auxiliary loss term to the performance of the MHCAC module, we perform an

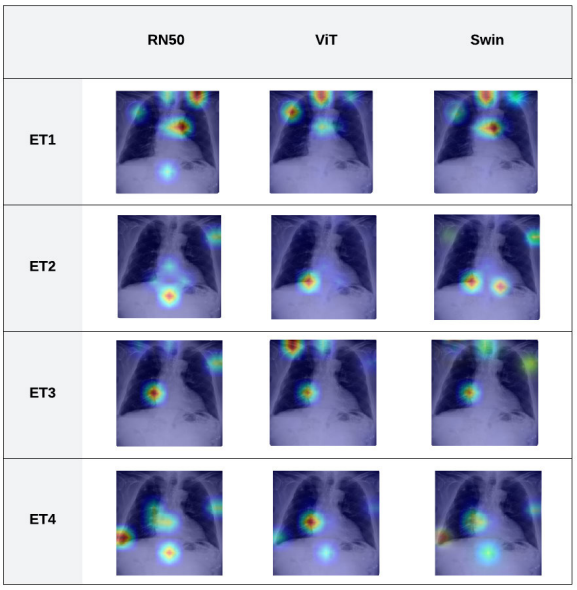


FIGURE 8. Comparison of attention heatmaps for a given expert token (ET) across RN50, ViT, and Swin encoders, highlighting variations in attention distribution despite consistent region focus on the MIMIC-CXR test set.

TABLE 7. Ablation study on the contribution of loss functions in the MHCAC module during the pretraining stage using the MIMIC-CXR training set. Mean F1 score is calculated across 5 common pathologies on the MIMIC-CXR test set.

Loss Functions				Mean F1 Score
CLS	OL	SL	CL	
✓	–	–	–	0.551
✓	✓	–	–	0.649
✓	✓	✓	–	0.667
✓	✓	✓	✓	**0.701**

ablation study by progressively adding Orthogonality Loss (OL), Sparsity Loss (SL), and Contrastive Loss (CL), while keeping the Classification Loss (CLS) fixed at training time.

- **Cross Entropy Loss(CLS)**: Provides the base supervision signal for abnormality classification.
- **Orthogonality Loss(OL)**: Enforces diversity among expert token representations by penalizing redundancy via orthogonality constraints. This ensures that each expert token specializes in capturing a unique subset of abnormality-specific features.
- **Sparsity Loss(SL)**: Promotes localized and interpretable attention by encouraging expert tokens to attend to the most salient regions in the image, thus reducing noisy or diffuse focus.
- **Contrastive Loss(CL)**: Enhances discriminability by explicitly increasing the inter-class separation and managing uncertain cases. It sharpens class decision boundaries through feature space regularization.

As observed from Table 7, the performance gains are consistent and cumulative as each auxiliary loss is introduced. While the classification loss (CLS) establishes the fundamental supervision signal, the inclusion of OL, SL, and CL sequentially reinforces representation diversity, spatial selectivity, and semantic separation, respectively. These loss functions act in a complementary fashion: OL prevents redundancy in token features, SL sharpens attention localization, and CL structures the embedding space to better handle class overlaps and uncertainty. Their synergy strengthens both the discriminative and explanatory capabilities of the model, culminating in improved F1 scores and report generation quality.

3) EFFECT OF INSTRUCTION-TUNED LLM

This ablation study compares the performance of using a base LLM, a fine-tuned LLM, and an instruction-tuned LLM for radiology report generation. We use Vicuna 7B [75] LLM for this experiment. The aim is to assess how the fine-tuning and instruction tuning of the LLM in radiology-specific data improve the quality of the generated text, focusing on clinical relevance and accuracy, evaluated through BERTScore, CIDEr, BLEU-4, ROUGE-L, and METEOR metrics.

A base LLM, trained on general-purpose datasets, lacks the domain-specific knowledge required for generating accurate radiology reports. This leads to vague or imprecise descriptions and fails to capture critical medical findings, reflected in lower scores. Additionally, the lack of understanding of the structured nature of radiology reports results in poorer performance on metrics like BLEU-4 and ROUGE-L, as the generated reports show less alignment with expert-written references.

In contrast, a fine-tuned LLM trained on radiology-specific data significantly improves performance. It accurately uses medical terminology and adheres to the correct report structure, leading to higher CIDEr, BLEU-4, ROUGE-L, METEOR, and BERT scores. The fine-tuned model produces more clinically relevant, coherent, and structured reports, closely aligning with the reference reports.

Conversely, as depicted in Figure 9, the instruction-tuned model, optimized through our prompt construction method, demonstrates superior performance as it has been specifically fine-tuned for META-CXR prompts that incorporate uncertainty class information. This underscores the effectiveness of instruction tuning in aligning the model with the task objectives, ensuring improved adaptability and performance in radiology report generation.

4) EFFECT OF COMMON EXPERT TOKENS

This ablation study investigates the impact of varying the number of common expert tokens on the classification performance, measured by weighted Precision, Recall, and F1-Score. Common expert tokens are specialized representations that extract high-level features relevant to abnormalities in radiographs. As the number of common expert tokens

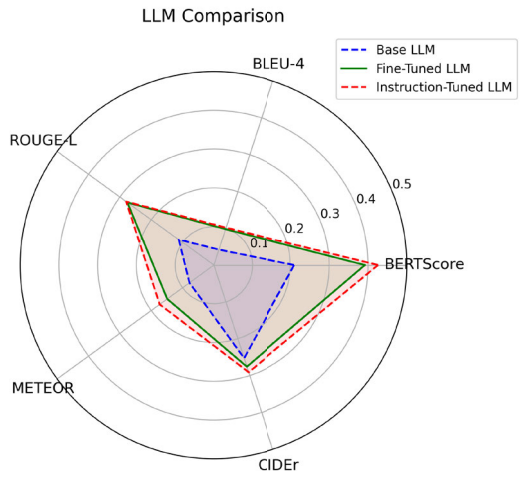


FIGURE 9. Effect of Tuned LLMs.

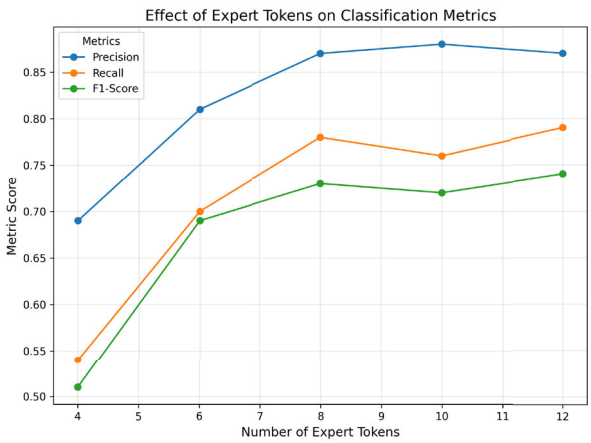


FIGURE 10. Effect of Expert Tokens in MHCAC.

increases, the extracted features adapt to provide a more comprehensive representation of abnormalities.

As depicted in figure 10 the results indicate that using 8 expert tokens achieves the optimal balance between Precision (0.87), Recall (0.78), and F1-Score (0.73). At lower token counts (e.g., 4 and 6), the extracted features may lack sufficient granularity, leading to reduced performance, particularly in Recall (0.54 and 0.70, respectively). Conversely, increasing the token count beyond 8 (e.g., 10 and 12 tokens) does not significantly improve performance. While there are slight fluctuations in metrics, the diminishing returns suggest that additional tokens do not provide meaningful new information and may introduce redundancy.

We conclude that 8 expert tokens are ideal for capturing abnormality-specific high-level features, providing the best trade-off among classification metrics. This highlights the importance of tuning the number of expert tokens to optimize feature extraction for the radiograph abnormality detection task.

V. DISCUSSION

A. COMPARATIVE ANALYSIS OF MEDICAL VLMS

Table 8 provides a comparative evaluation of various medical vision-language models (VLMs) on the MIMIC-CXR test dataset. The results highlight three key advantages of META-CXR over existing methods: enhanced visual representation through multi-encoder fusion, improved knowledge integration without reliance on external sources, and uncertainty-aware abnormality classification.

1) SUPERIORITY OF META-FORMER OVER Q-FORMER

Models such as Ra-Dialog, which utilize Q-Former, depend on a single vision encoder (e.g., RN50), limiting their ability to capture diverse radiological features. In contrast, META-CXR employs META-Former, which integrates three complementary vision encoders (RN50, ViT, and Swin Transformer), enhancing feature extraction and representation. This architectural advantage contributes to a higher BERTScore (0.426) compared to Ra-Dialog (0.400), indicating improved semantic alignment between generated and ground-truth reports.

2) ELIMINATING THE NEED FOR EXTERNAL KNOWLEDGE BASES

KARGEN incorporates a knowledge graph (KG) and disease extractors to enhance medical knowledge integration. However, maintaining and updating external knowledge bases is resource-intensive, requiring continuous curation and expansion to remain clinically relevant. META-CXR, on the other hand, bypasses this limitation by integrating structured abnormality classification through the MHCAC module. The BERTScore score improvement from 0.421 (KARGEN) to 0.426 (META-CXR) underscores that structured classification alone, without relying on a knowledge graph, leads to a more clinically meaningful report generation process.

3) ADVANTAGE OF MHCAC FOR UNCERTAINTY-AWARE CLASSIFICATION

Existing methods such as Ra-Dialog integrate a classification step but are limited to binary classification (positive/negative), ignoring the inherent uncertainty in radiology interpretation. In contrast, META-CXR's MHCAC module introduces uncertainty-aware multi-class classification, allowing the model to explicitly capture cases where radiological findings are ambiguous.

A key distinction of META-CXR is its ability to incorporate this uncertainty-aware abnormality classification into the report generation pipeline. Figure 6 demonstrates the impact of explicitly integrating uncertain class predictions when generating radiology reports. The columns labeled META-CXR (PN) and META-CXR (PNU) illustrate the difference

TABLE 8. Comparative analysis of medical vision-language models (VLMs) evaluated on the MIMIC-CXR test dataset. IT: Instruction Tuning, KG: Knowledge Graph integration, MHCAC: Multi-Head Cross-Attention Classifier.

Method	Components				BERTScore
	Vision Encoders	LLM	Alignment Module	Medical Knowledge Integration	
Flamingo-CXR [35]	NFNet-F6	Chinchilla	Perceiver Resampler	IT	–
R2GenGPT [72]	Swin	Llama2-7B	Visual Mapper	IT	0.415
Med-PaLM 12B [73]	ViT	PaLM-E	–	IT	–
RaDialog-RG [17]	RN50	Vicuna 7B	Q-Former	IT + Classifier	0.400
KARGEN [39]	Swin	Llama2-7B	Feature Fusion	IT + KG + Disease Extractor	0.421
META-CXR (Ours)	RN50 + ViT + Swin	Vicuna 7B	META-Former	IT + MHCAC	**0.426**

TABLE 9. Cost of success analysis for META-CXR architectural variants. Inference time (on RTX 4070 16GB) and VRAM usage indicate computational overhead.

Model Config	Parameters		Inf. Time (s)	VRAM (GB)	Metrics	
	Total	Trainable			Macro F1 Score(CE)	BERTScore
Baseline (RN50 + Q-Former) + Vicuna 7B	136M + 7B	108M	3.1	12.2	0.361	0.312
Multi-Enc. + META-Former + Vicuna 7B	255M + 7B	110M	3.3	13.5	0.378	0.324
+ MHCAC	295M + 7B	150M	3.5	14.8	0.389	0.392
+ LoRA Adapters (META-CXR)	300M + 7B	155M	3.5	14.9	**0.428**	**0.426**

between excluding and including uncertainty-aware classification in prompt construction. In the PN setting, which only considers positive and negative abnormalities, the generated reports lack explicit mention of uncertainty for specific conditions. For example, in the second row, the PN-generated report states, "The cardiomediastinal contour remains stable. Heart size is normal." without acknowledging any ambiguity regarding cardiomegaly. However, in the PNU setting, which incorporates the uncertain class, the corresponding report explicitly states, "The cardiomediastinal contour is stable. Heart size is within normal limits, though cardiomegaly remains uncertain." This added nuance improves the clinical relevance of the report by reflecting the inherent ambiguity in radiological assessments.

The results in Table 8 demonstrate that META-CXR achieves superior performance through multi-encoder integration (META-Former), structured abnormality classification (MHCAC), and instruction-tuned report generation without relying on external knowledge bases. These advancements establish META-CXR as a robust and clinically meaningful VLM for radiology applications.

B. COST OF SUCCESS ANALYSIS FOR META-CXR

Table 9 presents a step-by-step cost-benefit analysis of the META-CXR architecture, highlighting the impact of each module on performance and computational requirements. The baseline model, composed of a ResNet-50 encoder, Q-Former, and Vicuna-7B, provides the most lightweight configuration, with the lowest VRAM usage (12.2 GB) and inference time (3.1s), but also yields the lowest clinical efficacy and text generation performance.

Incorporating multi-encoder features along with the META-Former module moderately increases resource consumption while improving generation quality (BERTScore 0.324), validating the benefit of capturing complementary visual representations. The addition of the MHCAC module further enhances clinical performance, increasing the Macro F1-score to 0.389.

The full META-CXR configuration—augmented with instruction-tuned LLM components via LoRA adapters—achieves the highest scores across all metrics, reaching 0.426 BERTScore and 0.428 Clinical Efficacy F1. Importantly, this is accomplished with only a marginal increase in inference time (3.5s) and VRAM usage (14.9 GB).

It is worth noting that the total parameter count and memory footprint are primarily dominated by the Vicuna-7B language model. To ensure practical deployment, we use an `INT8 quantized` version of Vicuna-7B, allowing the entire model to run efficiently on a single RTX 4070 (16GB) GPU without degrading generation quality.

Overall, these results demonstrate that META-CXR offers strong performance improvements with only modest increases in computational cost, making it a scalable and deployable solution for real-world clinical settings.

C. OPTIMIZED THRESHOLDING FOR ENHANCED CLASSIFICATION PERFORMANCE

To improve classification accuracy at inference, we employ an optimized thresholding strategy that dynamically adjusts decision boundaries for each class. Instead of using a fixed threshold, we determine the optimal threshold by analyzing AUC-ROC curves and selecting the point closest to the top-left corner, representing the ideal classification trade-off between sensitivity and specificity. This selection is computed using the Euclidean distance formula as given

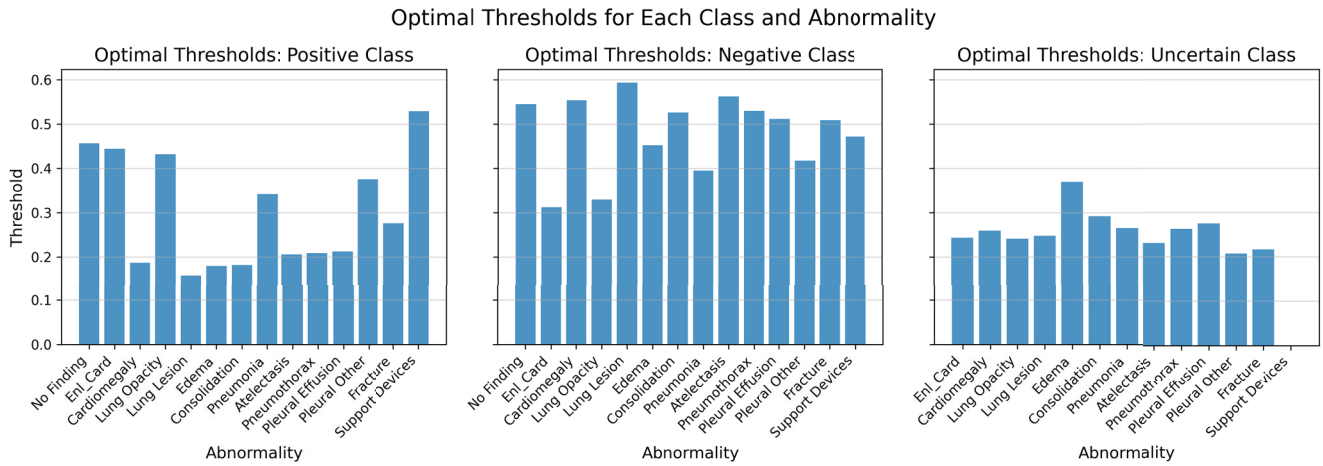


FIGURE 11. Classification Thresholds.

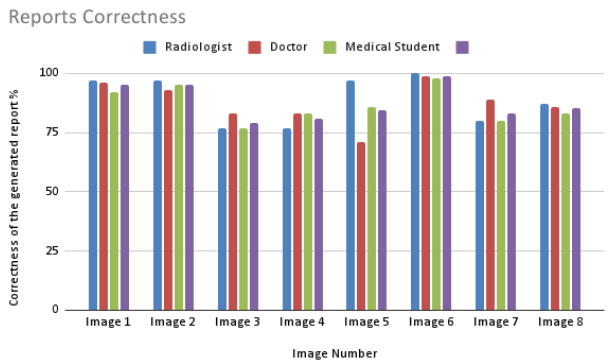


FIGURE 12. Experts Evaluation on Report Correctness.

in (22):

$$Distance = \sqrt{(1 - TPR)^2 + (FPR)^2} \tag{22}$$

The threshold corresponding to the minimum distance is chosen as optimal. As shown in Figure 11, this method effectively refines decision boundaries across positive, negative, and uncertain classes, reducing misclassifications and improving reliability.

By tailoring thresholds per class, META-CXR ensures that uncertain cases are neither ignored nor misclassified as positive or negative, a crucial factor in clinical decision-making. This uncertainty-aware thresholding not only enhances classification precision but also improves report generation by refining the soft prompts used by the LLM.

D. EXPERT VALIDATION OF REPORT GENERATION

To assess the clinical reliability and practical applicability of META-CXR, we conducted an expert validation study to evaluate the accuracy of generated reports and the interpretability of attention maps. A total of eight frontal chest radiographs covering diverse pathological cases were reviewed by domain experts. Ratings were collected on a scale of 1 to 5, where 5 indicated the highest correctness. Additionally, qualitative feedback was gathered to identify potential modifications to the generated reports and to assess the value of computational tools in radiology workflows.

The study included three board-certified radiologists with over five years of experience, seven physicians with three to five years of experience, and ten medical interns with at least one year of experience in chest X-ray interpretation. The primary objective was to measure the correctness of the generated reports and determine the effectiveness of explainability mechanisms in aiding clinical decision-making.

Figure 12 presents the average expert ratings for both report correctness and attention map interpretability. The correctness of the generated reports was rated at 88.75% by radiologists, 87.32% by physicians, and 86.75% by medical interns. Similarly, attention map interpretability scores were 88.75% for radiologists, 80.17% for physicians, and 78.87% for medical interns.

These results indicate that META-CXR effectively produces clinically relevant reports, with high agreement among radiologists and physicians. The explainability features were well-received, reinforcing the model's potential for aiding clinical interpretation. The study further highlights the utility of META-CXR as both a decision-support system for radiologists and a learning tool for medical trainees. The feedback obtained suggests that integrating such AI-assisted reporting systems could streamline clinical workflows while maintaining interpretability and diagnostic reliability.

VI. CONCLUSION

In this work, we propose META-CXR, a vision-language model for chest X-ray report generation and abnormality classification. The model leverages multiple complementary vision encoders—ResNet, ViT, and Swin Transformer—for enriched visual feature extraction. These features are utilized by the META-Former module to enable unified vision-language representation learning. The MHCAC module introduces a hierarchical feature extraction mechanism,

enhancing abnormality classification performance and guiding the report generation process.

META-CXR achieves state-of-the-art results across both standard and clinical NLP metrics, including BERTScore and Clinical Efficacy F1-score, and attains a mean F1-score of 0.699 in cross-domain abnormality classification. Comprehensive ablation studies confirm the contribution of each architectural component—multi-encoder setup, META-Former, and MHCAC—to the overall performance of the model. Beyond quantitative evaluations, the effectiveness of META-CXR was further validated through an expert study involving clinical professionals. Their feedback highlighted the clinical relevance and interpretability of our approach, confirming its potential as an assistive tool for radiologists in automated report generation.

While META-CXR achieved strong results, several limitations remain to be addressed in future work. These include integrating conversational medical agents, adopting lightweight and domain-specific LLMs, incorporating more pathological knowledge for improved contextual understanding, and further optimizing the META-Former and MHCAC modules to reduce computational costs. Although our current multi-encoder setup enhances representation diversity, it introduces additional inference-time complexity. To mitigate this, we plan to explore replacing Transformer-based encoders (ViT and Swin) with more efficient architectures such as *Mamba* [76], a recently proposed state space model that offers linear-time sequence modeling with reduced memory and compute requirements. Moreover, a mixed-method strategy can be followed, by combining chest X-ray images with additional patient clinical data and textual information to create more accurate and clinically relevant radiology reports. Additionally, we will explore advanced explainability techniques, expand to multi-view and longitudinal imaging data, and incorporate more specialized abnormality-related content to further enhance diagnostic depth and clinical relevance.

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